

insight ■ workshop

Managing Backups

Once you start taking backups on a regular basis, the multiple copies become confusing to manage and deciding on a rotation scheme to reuse the backup media is essential. Several backup strategies have evolved over the years. The simplest scheme is to have two or more media—hard disks or CDs—and rotate them every day. For example, if you have two CD-RWs, you could use CD1 on Monday, CD2 on Tuesday, again CD1 on Wednesday, and so on. This is fine if you don't have much data, but doing a full backup everyday of hundreds of megabytes of data (on a 10 GB+ hard disk) gets tedious. Also, if you do this, you don't have any backups older than a day. If your files got infected by a virus a week ago, you wouldn't be able to use any of your backups.

A commonly used scheme called the Grandfather-Father-Son scheme is efficient and easy to implement. Here, you maintain three levels of backup files. This uses many more CDs, but is a good compromise between reliability and speed. To illustrate this strategy, let's work with a period of three months, where you take backups six days a week (Monday to Saturday). Thus, you would have a separate hard disk just for taking backups, where you would take daily incremental backups. Every weekend, you would take a full backup on a CD-RW. Maintain at least two such CD-RWs, though it would be better to have four, one for each week of the month and reuse the discs every month instead of every fortnight. Finally, you would take a full backup every month on a CD-R, which will last you for a very long time.

With professional backup software such as Veritas Backup Exec (www.veritas.com), such strategies can be fully automated. For home users and smaller organisations, it is not too tedious to do this manually, since the quantum of data would also be comparably very less.

maintains several versions of the backups, making it very useful when working with important documents. The software may detect even a corrupted file as a modified one and take a backup, but you can always amend that by using an earlier backup.

Once you run SecureDat, click **File > New entry** and select the folder to be monitored under the Source tab. Choose to include subfolders if you want. Under the Target tab, click 'Other backup directory' and select where the files should be mirrored. Under Options you can set how many versions of the backups to maintain and on what criteria the backups should be made. You can enable encryption for security and compression for space under the Advanced tab.

Lifeboat or rubber tube?

When taking backups, the kind of media you choose is very important. You would want a storage solution that can be relied on for many years. How you choose your media depends on the amount of data you need to backup, its importance and of course, your budget.

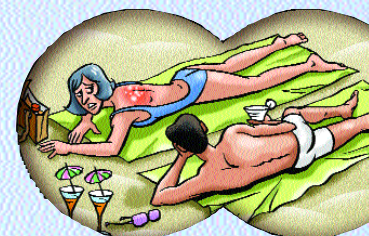
Every media type has dramatically different characteristics such as life span, storage capacity, re-usability and cost. Tape drives are the standard in the backup systems of many businesses. Tapes offer extremely high capacities, average data transfer rates, and a reasonable life span, but allow only sequential access to data.

For small businesses or home users, hard disks, CD-Rs, CD-RWs and even DVD-Rs would be a more viable backup option. The real advantage of a CD-Rs, CD-RWs and DVD-Rs is their long life spans—if maintained well, these are estimated to last as long as 30 years before oxidation starts affecting them.

Another option is to use removable drives. Zip drives and other external storage solutions (see, 'Walking Gigabytes' on page

72) have become quite popular in years. However, these media do not offer extremely high levels of reliability and are best suited for shuttling data between locations, occasional ad-hoc backups and short-term storage.

For an optimum balance between reliability and performance, you should use a combination of storage solutions for different kinds of backups (see box, 'Managing backups'). Thus, you can use a hard disk for daily backups, CD-RWs for weekly backups and CD-Rs for monthly backups. You can even use just a hard disk for all backups. It's prudent to keep the originals of the backups in different places, in case a hard disk fails and you lose both.



Jumping back

As with backups, there are a few cardinal rules to restoring, too. One of the rules of taking backups is to check the integrity of each backup on a regular basis. If you do not do this regularly, you could be in for a shock when you need to restore some files. As with your primary storage media, backup media can also get corrupted—CD-Rs can get scratched, hard disks ma-



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